

Backup slides

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Backup: budget configuration

Parameter	Value	Enforced in
max_depth	4	both routers
beam_width	3	explorer (slice)
max_anchors	5	explorer <i>and</i> evaluator
max_backtracks	3	post-evaluation router
max_llm_calls	25	LLM client
max_verify_iters	2	post-verification router, verifier
max_seconds	300	both routers

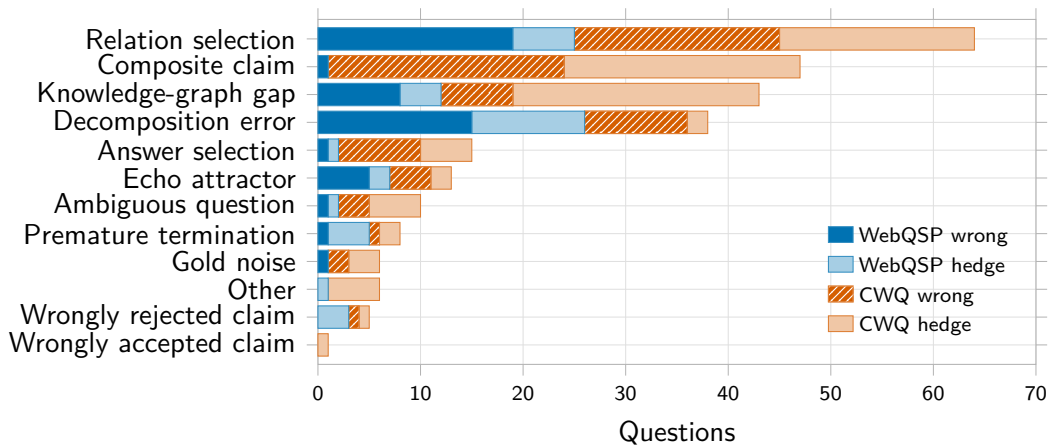
Termination does not depend on model behaviour. Every cycle passes through a router that checks a monotone counter.

Backup: which budgets actually bind

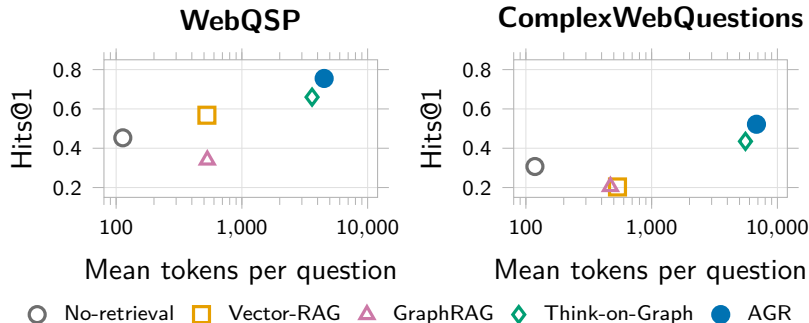
Budget	WebQSP	CWQ	Pooled
Depth cap	16.5%	34.8%	25.6%
Backtrack cap	3.2%	9.2%	6.2%
Verify-iteration cap	1.0%	2.8%	1.9%
Call cap	0.0%	0.0%	0.0%

Percentage of questions on which each budget refused a request. AGR never reaches the 25-call cap. That is what makes a clipped Think-on-Graph a meaningful comparison.

Backup: the full failure census



Backup: accuracy against cost



On *tokens* Think-on-Graph is **not** dominated. It buys lower accuracy at a lower token price. On *calls*, which the budget meters, it is.

The frontier depends on which cost you charge.

Backup: the benchmark was wrong 57 times

The same cross-system agreement is also a **detector**. Consensus flagged 105 questions and adjudication confirmed 41. The gap is the attractor above, not label noise.

- 41 excluded before the census
- 16 more found inside it, as gold-noise or ambiguous rows
- No question counted in both

⇒ 57 **distinct questions**

Why report it

Reported defect rates for these two standard benchmarks are rare and usually anecdotal. This is a counted rate with a documented adjudication procedure and per-question provenance.

A contribution that outlives this particular system.

Backup: what changed from the proposal

Proposal	Now	Why
“MCTS-inspired” search	Withdrawn	No rollouts and no value backpropagation. It is guided best-first search with a bounded beam and backtracking.
FactBench as graph and judge	Dropped	An evaluator whose facts sit inside the retrieval graph cannot measure hallucination.
Path fidelity metric	Deferred	Needs gold SPARQL relation chains, which the RoG distribution does not carry.
A registered protocol	Pre-specified	Four decisions were fixed in writing, but nothing was filed with a registry.

Backup: what comes next

Repairs the census earned

- Tell the decomposer which argument is the answer (9 extraction-bug cases)
- Log accepted claims with the triples that matched them
- Re-run the baselines at equal candidate widths and a wider radius

Operators the failures specify

- Gate decomposition on question structure
- A set-intersection operator (47 composite-claim cases)
- Ordinals and literal values (43 knowledge-graph-gap cases)

Fact verification against a knowledge graph is the easier half. Relevance verification is the half still open.